

Functional Annotation of *ilvI* (Q88DY8, PP_4680) in *Pseudomonas putida* KT2440

1. Summary / Answer

Q88DY8 (gene *ilvI*, locus PP_4680) is the large, catalytic subunit of **acetoxyacid synthase (AHAS; also called acetolactate synthase, EC 2.2.1.6)** — a thiamine-diphosphate (ThDP)-dependent enzyme that catalyzes the **first and committed step of branched-chain amino acid (BCAA) biosynthesis** (valine, leucine, isoleucine). Acting in the **cytoplasm** as part of an A₂B₂ holoenzyme with a small regulatory subunit (PP_4679/*ilvH*), it condenses two molecules of pyruvate into (2*S*)-2-acetolactate (the Val/Leu precursor), or condenses pyruvate with 2-ketobutyrate into 2-aceto-2-hydroxybutyrate (the Ile precursor), releasing CO₂. The reaction requires ThDP, a divalent metal (Mg²⁺), and FAD.

Gene-identity verification confirms this is the correct protein: UniProt, KEGG, GO, and InterPro all independently annotate Q88DY8/PP_4680 as the AHAS large subunit, and the surrounding literature on the AHAS enzyme family is fully consistent with the provided protein description, domains, and organism.

2. Gene / Protein Identity Verification (mandatory check)

Attribute	Provided (UniProt)	Independently confirmed
Accession	Q88DY8	UniProt REST ✓
Protein	Acetolactate synthase, EC 2.2.1.6	UniProt recommended name; KEGG "acetoxybutanoate synthase / acetolactate synthase" ✓
Gene / locus	<i>ilvI</i> / PP_4680	KEGG ppu:PP_4680 ✓
Organism	<i>P. putida</i> KT2440	✓
Family / domains	TPP enzyme family; AHAS_TPP, ThDP-binding, DHS-like NAD/FAD-binding	UniProt SIMILARITY + KEGG orthology K01652 ✓
KEGG ortholog	—	K01652 = "acetolactate synthase I/II/III large subunit [EC:2.2.1.6]" ✓
Length	—	574 aa

Note on nomenclature: In enterobacteria (*E. coli*), there are three AHAS isozymes whose large catalytic subunits are named *ilvB* (AHAS I), *ilvG* (AHAS II), and *ilvI* (AHAS III); each pairs with a small regulatory subunit (*ilvN*, *ilvM*, *ilvH*, respectively) (Vyazmensky et al., 2009,).

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The *P. putida* gene retains the *ilvI* name for its single AHAS large subunit. **The symbol is therefore unambiguous here and the literature on AHAS applies directly.** This is *not* the same as, e.g., unrelated genes that share the "ilvI" text in other contexts.

3. Primary Function: Reaction and Substrate Specificity

AHAS catalyzes a ThDP-dependent carbonylation. UniProt records the canonical reaction (RHEA:25249):



The enzyme has two physiological condensations that partition flux between the BCAA branches (Liu, Li & Wang, 2016,

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- **Pyruvate + pyruvate → (2S)-2-acetolactate** → downstream to **L-valine** and **L-leucine**.
- **Pyruvate + 2-ketobutyrate → 2-aceto-2-hydroxybutyrate** → downstream to **L-isoleucine**.

Catalytic mechanism. ThDP (activated by the divalent metal ion, Mg^{2+}) is deprotonated at C2 to a reactive ylide; it attacks the first pyruvate and, after decarboxylation (loss of CO_2), forms a hydroxyethyl-ThDP (enamine/carbanion) intermediate. This carbanion then attacks the carbonyl of a second 2-oxoacid (pyruvate or 2-ketobutyrate), and product release regenerates the ylide (Gokhale & Tilak, 2015,

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Cofactors (UniProt COFACTOR; GO): **ThDP** (1 per subunit), **Mg^{2+}** (1 per subunit), and **FAD**. FAD is a non-catalytic, structurally required cofactor — a vestige of AHAS's evolutionary descent from the FAD/redox-dependent pyruvate oxidase family (Gokhale & Tilak, 2015,

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Substrate specificity (from a close bacterial homolog). Kinetic characterization of the orthologous *Ralstonia eutropha* AHAS (ilvBH) shows the enzyme has weak affinity for the first substrate pyruvate ($K_m \approx 10.5 \mu M$) but strong selectivity toward **2-ketobutyrate as the second substrate (specificity ratio $R \approx 140$)**, explaining efficient isoleucine synthesis when 2-ketobutyrate is available (Lu et al., 2015,

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While measured in *R. eutropha*, this behavior is characteristic of the conserved bacterial AHAS catalytic subunit to which PP_4680 belongs.

Substrate specificity inferred from conserved active-site residues. A more ortholog-specific inference comes from the active site itself. *E. coli* AHAS isozyme I (IlvB) is unusual in being **non-specific** for 2-ketobutyrate versus pyruvate as the acceptor substrate; this is caused by

two residues (Gln480 and Met476) that replace a **Trp and Leu conserved in the other AHAS isozymes**, accelerating the ligation/product-release steps (Belenky et al., 2012,

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Aligning Q88DY8 to these positions shows that ***P. putida* IlvI retains the conserved Trp (Trp486)** — exactly as in *E. coli* AHAS III (Trp484) — rather than the AHAS I Gln. It therefore mirrors the AHAS III active site and is predicted to be **specific for 2-ketobutyrate over pyruvate as the second substrate**, i.e., an efficient producer of the isoleucine precursor. This structure/evolution-based prediction is more reliable than generic family-level transfer because it is anchored on the specific catalytic residues of the 67%-identical AHAS III ortholog.

4. Quaternary Structure, Regulatory Subunit, and Localization

Holoenzyme architecture. Active AHAS is a **holoenzyme of a catalytic (large) subunit + a regulatory (small) subunit**. The crystal structure of the AHAS holoenzyme shows a **A₂B₂ arrangement** (two catalytic + two regulatory subunits), with the regulatory-subunit dimer clamping the catalytic dimer; a conserved salt bridge acts as a trigger that opens the catalytic activation loop, boosting catalytic-subunit activity (Zhang et al., 2020,

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The isolated large subunit has only a small fraction of full activity, and holoenzyme activity is fully restored upon reconstitution with the small subunit (Vyazmensky et al., 2009,

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The partner small subunit in *P. putida*. Chromosomal context strongly corroborates this: PP_4680 (*ilvI*, K01652, large subunit) is immediately adjacent to **PP_4679 (K01653, "acetolactate synthase small subunit" = *ilvH*/regulatory subunit)**, and directly upstream lies **PP_4678 (K00053, ketol-acid reductoisomerase / *ilvC*)**, the enzyme catalyzing the *next* step of the pathway. This ***ilvC*–*ilvH*–*ilvI* clustering** is typical of co-regulated BCAA-biosynthesis genes and confirms the functional pairing of PP_4680 with PP_4679 (KEGG genome positions: PP_4680 5317649–5319373; PP_4679 5317156–5317647, same strand).

Allosteric feedback inhibition. The regulatory subunit carries paired **ACT domains**; the end-product amino acid(s) **valine (and/or leucine) bind at the interface between two ACT domains**, transmitting inhibition to the catalytic subunit (Barak & Chipman, 2012,

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This makes AHAS the **rate-controlling gatekeeper** of the pathway: BCAA end-products shut down their own synthesis at the first committed step. Homolog kinetics quantify the effect (*R. eutropha* AHAS: L-valine $IC_{50} \approx 1.2$ mM, L-isoleucine ≈ 2.3 mM, L-leucine ≈ 5.4 mM; Lu et al., 2015,

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Barak & Chipman also note that the regulatory subunits of "typical" single AHAS enzymes in non-enterobacteria resemble *E. coli* AHAS III (ilvIH) — consistent with the *P. putida* ilvIH-type system.

Subcellular localization. AHAS is a **soluble cytoplasmic enzyme** (GO:0005948, "acetolactate synthase complex"; no signal peptide or transmembrane region). Its function — a biosynthetic condensation feeding downstream cytoplasmic enzymes (ilvC, ilvD, and the transaminases/leucine-branch enzymes) — is carried out entirely within the cytoplasm.

PP_4680 is the dedicated biosynthetic AHAS. *P. putida* KT2440 encodes four TPP-family proteins that KEGG maps to the AHAS-large-subunit ortholog K01652 (PP_1157, PP_1394, PP_3365, PP_4680). However, several lines of evidence single out **PP_4680** as the true anabolic BCAA enzyme: (i) the genome contains **only one small regulatory-subunit gene, PP_4679 (K01653)**, and it sits **immediately adjacent to PP_4680** in the ilvC-ilvH-ilvI cluster; (ii) PP_4680 is uniquely annotated "**acetohydroxybutanoate synthase / acetolactate synthase**," i.e., competent for the 2-ketobutyrate→isoleucine branch that defines the biosynthetic enzyme; and (iii) the other three loci are standalone TPP enzymes with no cognate small subunit or *ilv*-operon context (likely catabolic acetolactate/2-oxoacid-decarboxylase-type activities). At the sequence level, Q88DY8 carries the canonical ThDP-binding fingerprint **GDG...N** (...YDKGDGNVIKP..., GDG at residue 369), the hallmark motif that coordinates the ThDP-Mg²⁺ cofactor in this enzyme family. Together these confirm PP_4680/*ilvI* as the correct, dedicated biosynthetic AHAS catalytic subunit.

Orthology confirmed by sequence identity. Global (Needleman–Wunsch) alignment shows Q88DY8 (574 aa) is **67.0% identical to the *E. coli* AHAS isozyme III large subunit (ilvI, P00893; also 574 aa)** — versus only **44.8% to the AHAS I large subunit (ilvB, P08142)**. This markedly higher identity to *E. coli* AHAS III, plus the identical chain length, establishes Q88DY8 as a **specific ortholog of the AHAS III (ilvIH) lineage** rather than a generic AHAS. This is fully consistent with the observation that "typical" single bacterial AHAS enzymes resemble *E. coli*

AHAS III (Barak & Chipman, 2012, [P 22198286](#)), and it independently validates the *ilvI* gene name from evolutionary evidence.

5. Pathway Context

AHAS catalyzes **step 1 of 4** in two parallel routes (UniProt PATHWAY; KEGG ppu00290 "Valine, leucine and isoleucine biosynthesis"):

- **L-valine from pyruvate:** step 1/4 → acetolactate → (ilvC) → (ilvD) → 2-ketoisovalerate → (ilvE/transaminase) → valine. 2-Ketoisovalerate also branches to **leucine** (leuABCD) and is a precursor for **pantothenate/CoA**.
- **L-isoleucine from 2-oxobutanoate (2-ketobutyrate):** step 1/4 → 2-aceto-2-hydroxybutyrate → (ilvC) → (ilvD) → 2-keto-3-methylvalerate → (ilvE) → isoleucine. (2-Ketobutyrate itself derives from threonine via threonine dehydratase, *ilvA*.)

Because it initiates all three BCAAs and their derived cofactor precursors, and because the pathway is **absent in animals**, AHAS is the well-known molecular target of >50 commercial herbicides (sulfonylureas, imidazolinones, triazolopyrimidines) and an antimicrobial/anti-tuberculosis drug target (Liu et al., 2016, [P 27576495](#));

Gokhale & Tilak, 2015, [P 25882218](#));

Lonhienne et al., 2018, [P 29440497](#)).

6. Hypotheses: Supported and Refuted

Supported 1. *ilvI*/PP_4680 encodes the AHAS **large catalytic subunit** — supported by UniProt, KEGG K01652, GO, InterPro, and direct concordance with AHAS literature. 2. It catalyzes ThDP/Mg²⁺/FAD-dependent condensation of pyruvate (±2-ketobutyrate) as the committed step of

BCAA biosynthesis — supported (UniProt reaction,

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3. It functions as a **cytoplasmic holoenzyme** with the adjacent small subunit PP_4679 (*ilvH*) and is feedback-inhibited by valine via ACT-domain sites — supported by structural/kinetic literature

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and by the *P. putida* *ilvC*–*ilvH*–*ilvI* gene cluster.

Refuted / ruled out - That the *ilvI* symbol refers to an unrelated non-AHAS gene — refuted; all databases and family/domain evidence converge on AHAS. - That the large subunit alone constitutes the physiologically regulated enzyme — refuted; full activity and valine sensitivity require the small subunit

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7. Evidence Quality and Limitations

- **Enzyme identity, reaction, cofactors, pathway, and localization** rest on strong, convergent database annotation (UniProt/KEGG/GO/InterPro) plus authoritative reviews and primary structural studies of the AHAS family — high confidence.
- **Direct biochemical characterization of the *P. putida* KT2440 enzyme specifically was not located** in the searchable literature; the quantitative substrate-specificity and feedback-inhibition parameters cited (K_m , IC_{50} , selectivity R) come from a **close bacterial homolog** (*R. eutropha* *ilvBH*,

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and from *E. coli* isozyme studies. These are transferable by strong homology but are inferences, not measurements on Q88DY8.

- **Future directions:** heterologous expression and kinetic assay of PP_4680 + PP_4679 to measure its own K_m/k_{cat} and valine IC_{50} ; verification that PP_4679 is the cognate regulatory subunit by co-purification; and testing whether *P. putida* encodes any second AHAS large subunit (an *ilvBN*-type paralog) elsewhere on the chromosome.

References (PMIDs)

- Vyazmensky et al., 2009 — Interactions between large and small AHAS subunits
([P 19653643](#))
- Liu, Li & Wang, 2016 — AHAS evolution, structure, and function (review)
([P 27576495](#))
- Gokhale & Tilak, 2015 — Mechanisms of bacterial AHAS; drug target
([P 25882218](#))
- Zhang et al., 2020 — Molecular architecture of the AHAS holoenzyme (A_2B_2)
([P 32538427](#))
- Barak & Chipman, 2012 — Allosteric regulation / ACT-domain feedback sites
([P 22198286](#))
- Lu et al., 2015 — Kinetics of *R. eutropha* AHAS (substrate specificity, feedback)
([P 25081555](#))
- Belenky et al., 2012 — Active-site residues controlling AHAS I vs III specificity/kinetics
([P 22443469](#))
- Lonhienne et al., 2018 — Structural basis of AHAS herbicide inhibition
([P 29440497](#))

- Platko, Willins & Calvo, 1990 — *ilvIH* operon regulation in *E. coli*

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- Database sources: UniProt Q88DY8; KEGG ppu:PP_4680 / K01652 / ppu00290; InterPro IPR012846, IPR039368, IPR029035, IPR029061, IPR012000.

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